

## Chapter 4

# The Motivation to Elaborate in a Relatively Objective Manner

### Introduction

We have now discussed some of the major variables that can affect a person's *ability* to scrutinize issue-relevant arguments in a relatively objective manner. Motivational variables are also important in affecting the elaboration likelihood. If a person is highly able to process a message but lacks the requisite motivation, little processing will occur. In this chapter we first discuss three situational variables that exert an important effect on motivation to process—the personal relevance of an issue, personal responsibility for message evaluation, and the number of message sources. Then, we discuss an individual difference variable we have developed that assesses motivation to think—the need for cognition.

### Factors Affecting the Motivation to Elaborate Objectively

#### Effects of Personal Relevance on Message Elaboration and Persuasion

Perhaps the most important variable affecting the motivation to process a persuasive message is the personal relevance of the advocacy. Previous social psychological analyses of personal relevance have labeled this construct (or variations of it) "ego-involvement" (Rhine & Severance, 1970; Sherif, Sherif, & Nebergall, 1965), "issue involvement" (Kiesler, Collins, & Miller, 1969), "personal involvement" (e.g., Apsler & Sears, 1968; Sherif, Kelly, Rodgers, Sarup, & Tittler, 1973), "vested interest" (Sivacek & Crano, 1982), and others. In brief, we regard personal relevance as the extent to which an advocacy has "intrinsic importance" (Sherif & Hovland, 1961) or "personal meaning" (Sherif et al., 1973). Personal relevance occurs when people expect the issue "to have significant consequences for their own lives" (Apsler & Sears, 1968). Of course, relevance can be judged in terms of

a variety of dimensions such as the number of personal consequences of an issue, the magnitude of the consequences, and their duration. For example, some advocacies may remain high in personal relevance for many people over a long period of time (e.g., changing the U.S. income tax structure), other advocacies may have personal relevance for a more circumscribed period and/or audience (e.g., raising college tuition), and still other advocacies may have personal relevance only under certain very transient conditions (e.g., refrigerator ads have higher relevance when a person is in the market for this appliance).

Most of the early research on the personal relevance of an issue indicated that increasing personal involvement was associated with resistance to persuasion (Miller, 1965; Sherif & Hovland, 1961), and the most prominently mentioned explanation for this finding was derived from social judgment theory (Sherif, Sherif, & Nebergall, 1965). Involvement was believed to be associated with a greater probability of message rejection because people were postulated to hold expanded "latitudes of rejection" as personal involvement increased, and incoming messages would therefore be more likely to fall within the unacceptable range of a person's implicit attitude continuum (Eagly & Manis, 1966). To account for the fact that increasing relevance was associated with increased resistance mostly for counterattitudinal and not proattitudinal issues (e.g., Eagly, 1967), Pallak, Mueller, Dollar, and Pallak (1972) proposed that increasing involvement (or commitment) increased the probability of rejecting counterattitudinal messages because these messages were *contrasted* (seen as further away from one's own position than they really were and therefore more objectionable), but proattitudinal messages were assimilated (seen as closer to one's own position and therefore more acceptable).

Importantly, explanations of involvement based on social judgment theory did not consider the nature of the issue-relevant arguments presented in the communication. Instead, as involvement increased, some messages were thought to induce increased assimilation (and acceptance) or increased contrast (and rejection) based on the particular positions they were judged to espouse. We suggested an alternative analysis of the effects of personal involvement (Petty & Cacioppo, 1979b). Specifically, we proposed that as personal relevance increases, people become more motivated to process the issue-relevant arguments presented. As the personal consequences of an advocacy increase, it becomes more important for people to form a veridical opinion because the consequences of being incorrect are greater. Because of these greater personal consequences, people should be more motivated to engage in the cognitive work necessary to evaluate the true merits of the proposal.

Much of the early work on issue involvement was conducted by finding existing groups of people who differed in the extent to which an issue was important, and thus was correlational in nature (e.g., Hovland, Harvey, & Sherif, 1957). More recent investigators have chosen to study issue relevance

by varying the issue and message between subjects (e.g., Lastovicka & Gardner, 1979; Rhine & Severance, 1970). In other words, some undergraduate students might receive a message on a highly involving issue (e.g., instituting a new exam policy), whereas others would receive an issue of low relevance (e.g., raising import duties on raw silk). Although this research is interesting in that these involvement classifications probably capture the relevance concept as it occurs in the "real world," several interpretive problems are introduced. Specifically, distinctions based on different kinds of people or different issues may confound personal relevance with other factors (see discussion by Kiesler, Collins, & Miller, 1969). One particularly likely confound is that people in the high relevance groups or who receive the high relevance issues may be more familiar with the issue and may have more topic-relevant knowledge (e.g., Lutz, MacKenzie, & Belch, 1983; Wood, 1982). Thus, in addition to possessing greater motivation to process the messages, it is likely that these subjects also have greater ability to do so. Thus, when a message contains information that is inconsistent with subjects' initial opinions, high relevance subjects should be more motivated and generally more able to generate counterarguments to the arguments presented. However, when a message contains information that is consistent with the subjects' initial attitudes, high relevance subjects should be more motivated and generally more able to elaborate the strengths of the arguments. In sum, it is possible that differences in message-relevant elaboration between high and low relevance subjects may account for the different effects obtained for pro and counterattitudinal issues in previous research on personal involvement.

#### *Personal Relevance Can Enhance Message Elaboration*

In order to test our formulation, we first sought to replicate previous research using a manipulation of personal relevance that did not include differences in familiarity with the issue and arguments as a component (Petty & Cacioppo, 1979b, Experiment 1). Employing a procedure introduced by Apsler and Sears (1968), we had subjects in both high and low relevance groups receive the same message on the same topic, but high involvement subjects were led to believe that the advocacy would affect them personally, whereas low involvement subjects were led to believe that the advocacy would have no personally relevant implications. The message arguments were pretested so that the counterattitudinal message arguments were weak and elicited predominantly unfavorable thoughts and the proattitudinal message arguments were strong and elicited predominantly favorable thoughts when subjects were instructed to think about them. According to the enhancement of elaboration hypothesis, increased personal relevance should increase the motivation to process the message content. As a result of this, increased involvement should increase agreement with the proattitudinal (strong) advocacy and decrease agreement with the counterattitudinal (weak) advocacy.

**Method.** A total of 24 students participated in a 2 (Personal relevance: high or low)  $\times$  2 (Message type: strong proattitudinal or weak counterattitudinal) between-subjects factorial design. Upon arrival at the lab, subjects were told that students in a sound engineering course had prepared the communications that were to be employed in the investigation and that in return for the use of the tapes in other research, the investigators had agreed to obtain evaluations of the sound quality of the audiotapes. The subjects were asked to aid in this endeavor.

Subjects listened to either a strong proattitudinal message extolling the virtues of more lenient coed visitation hours on college campuses, or a weak counterattitudinal message contending that colleges should be more strict in their coed visitation policies. To manipulate personal relevance, half of the subjects were told that the speaker was advocating that the change in visitation hours be implemented at their own university, whereas the other half were told that the speaker advocated the change for a distant college. After the presentation of the message, subjects read: "Because your own opinion about the position advocated on the tape may influence the way you rate the quality of the tape, we would like to obtain a measure of how you feel about the views proposed by the speaker on each scale below." Subjects responded to attitude, thought-listing, and ancillary measures.

**Results.** Subjects rated the extent to which they found the communication "involving" on an 11-point scale where 11 indicated "extremely involving." Subjects in the high involvement conditions rated the message as significantly more involving than subjects in the low involvement condition. In addition, analyses of the listed thoughts indicated that subjects generated predominantly favorable thoughts to the proattitudinal communication, whereas they generated predominantly unfavorable thoughts to the counterattitudinal communication. These data provide support for the effectiveness of the experimental manipulations.

As depicted in Figure 4-1, a Message type  $\times$  Personal relevance

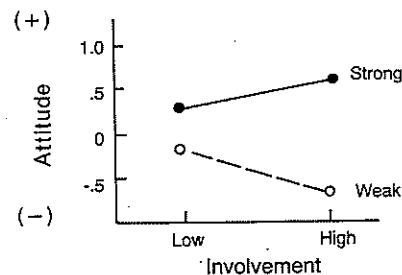


Figure 4-1. Postmessage attitudes as a function of the personal relevance of the issue and type of message (Proattitudinal/Strong or Counterattitudinal/Weak) (data from Petty & Cacioppo, 1979b, Experiment 1).

interaction was obtained on the measure of subjects' attitudes toward the change in visitation policy. When the message was counterattitudinal (and weak), increased relevance was associated with decreased acceptance, but when the message was proattitudinal (and strong), increased relevance was associated with greater acceptance. Similar Message type  $\times$  Relevance interactions on the thought-listing data provided further support for the elaboration-enhancement explanation of involvement effects: under high involvement, subjects generated more favorable thoughts and fewer unfavorable thoughts to the proattitudinal than counterattitudinal advocacy; under low involvement, however, neither the number of favorable thoughts nor unfavorable thoughts was affected by message type.

#### *A More Stringent Test of the Motivation to Elaborate Hypothesis*

Although our initial study on personal relevance provided evidence consistent with the view that increasing personal relevance enhances motivation to scrutinize message content, it is still possible that attitude change was mediated by assimilation/contrast effects (cf., Pallak, Mueller, Dollar, & Pallak, 1972), inasmuch as the strong arguments advocated a proattitudinal position and the weak arguments advocated a counterattitudinal one. To provide a stricter test of the view that personal relevance enhances message scrutiny, we conducted a second experiment (Petty & Cacioppo, 1979b, Experiment 2) in which all subjects were exposed to a counterattitudinal message advocating that college seniors should be required to pass a comprehensive exam in their major area as a requirement for graduation. For half of the subjects, the arguments in the message were strong and compelling, and for the other half, the arguments were weak and specious. Given that the messages advocated an identical counterattitudinal position, but differed in the quality of the arguments used to support that position, it becomes possible to evaluate various explanations of issue involvement. We expected that increased personal involvement would motivate subjects to process the information contained in the communication more carefully. Thus, even though high involvement may initially increase a subject's motivation to reject a counterattitudinal advocacy (or accept a proattitudinal one), subjects should ultimately show greater appreciation for the flaws in the weak communication and the virtues in the strong one as relevance increases.<sup>1</sup> The modified social judgment formulation, as outlined by Pallak et al. (1972), predicts that increased personal involvement will produce decreased persuasion for both messages, because each message adopts an identical position in opposition to the subjects.

<sup>1</sup>In short, we are suggesting that message position (pro or counterattitudinal) may bias message processing, whereas personal relevance per se affects information processing in a relatively objective manner. Of course, in any one persuasion context, unless a neutral advocacy is presented, both factors will determine the direction and intensity of message elaboration.

**Method.** In this study, 72 students participated in a 2 (Personal relevance: high or low)  $\times$  2 (Argument quality: strong or weak) between-subjects factorial design. Upon arrival at the lab, subjects were told that each year the psychology department assists the school of journalism in evaluating radio editorials that are sent in by colleges and universities throughout the country; their task would be to provide ratings of the quality of the editorials. Following these instructions, subjects heard the strong or weak 4-minute taped communication on senior comprehensive exams over headphones. For high relevance subjects, the editorial advocated that the exams be instituted at their own university, whereas for low relevance subjects, the editorial advocated that the exams be instituted at a distant university. After message exposure, subjects responded to attitude, thought-listing, recall, and ancillary measures.

**Results.** In order to determine whether our manipulation of involvement affected the amount of message processing in which subjects engaged, subjects were asked to rate on an 11-point scale how much thought they put into evaluating what the speaker had to say. Analyses revealed that subjects in the high involvement conditions reported doing more thinking about the messages than subjects in the low involvement conditions. These data support the view that increasing personal involvement affects the hypothesized mediating variable of message processing. The results on the attitude measure were similar to those in the preceding study (see Figure 4-2). An Argument quality  $\times$  Relevance interaction indicated that as relevance increased, subjects' attitudes showed greater discrimination of strong from weak arguments. More specifically, when the message was strong, increasing relevance produced a significant increase in attitudes, but when the message was weak, increasing relevance produced a significant

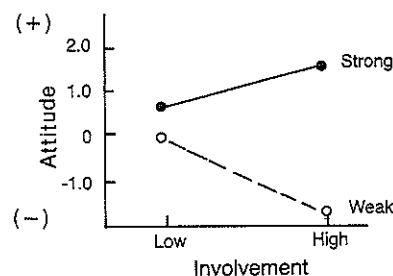


Figure 4-2. Postmessage attitudes as a function of the personal relevance of the issue and argument quality (data from Petty & Cacioppo, 1979b, Experiment 2).

decrease in attitudes. Similar Argument quality  $\times$  Relevance interactions on the thought-listing data also replicated the results of the preceding experiment: subjects generated more favorable thoughts and fewer unfavorable thoughts to the strong than weak arguments under high involvement, whereas neither the number of favorable thoughts nor unfavorable thoughts was affected by argument quality under low involvement.

Finally, there was no significant difference in argument recall between involvement conditions, although consistent with the message elaboration notion (cf., Craik & Lockhart, 1972), there was a slight tendency for subjects in the high involvement conditions to recall more arguments than subjects in the low involvement conditions ( $p < 0.08$ , one-tailed). Since message recall is an imperfect index of differences in message encoding (see Chapter 2), correlations were calculated separately for high and low involvement subjects between the measures of attitudes and issue-relevant thinking. The obtained pattern of correlations revealed that issue-relevant thinking was a better predictor of subjects' postcommunication attitudes under high than low relevance conditions. This has been found to be a replicable result in conceptually similar studies (Chaiken, 1980; Petty & Cacioppo, 1979a). Further analyses showed that the ability to recall message arguments did not allow any reliable prediction of postcommunication attitudes. Together, these data provide support for the view that increasing the personal relevance of a communication increases the importance of message elaboration in producing persuasion.

#### Reconsideration of Personal Relevance

In the context of examining the effects of other variables, we have replicated the interaction of personal relevance and argument quality several times (e.g., Petty, Cacioppo, & Heesacker, 1981; Petty, Cacioppo, & Schumann, 1983; Petty & Cacioppo, 1984a). These interactions provide support for the view that as the personal relevance of a message increases, people become more likely to undertake the cognitive work of evaluating the issue-relevant arguments presented. We have been able to show that this increased processing can result in people showing greater appreciation for the strengths of cogent arguments and the flaws in specious ones. Several cautions are in order, however, concerning the possible limitations of this effect. First, we suspect that there are some circumstances where personal interests are so intense, as when an issue is intimately associated with central values (e.g., Ostrom & Brock, 1968), that processing will either terminate in the interest of self-protection or will become biased in the service of one's own ego (e.g., Greenwald, 1980, 1981). Second, although we believe that personal relevance per se motivates people to seek correct attitudes, the direction of a message (pro or counterattitudinal) may more particularly motivate acceptance or rejection. All else equal, people would presumably prefer to accept messages that agree with their initial attitudes and reject messages that disagree with them. In our study, subjects were exposed to a

message advocating a change in university policy that, if implemented, might have prevented some students from obtaining their degrees. Although this advocacy was clearly counterattitudinal, it was not so threatening that the biased processing induced by the discrepancy overpowered the relatively objective processing of the strong arguments motivated by the high relevance of the communication. The joint consideration of personal relevance and message discrepancy suggests that as a message becomes more counterattitudinal, the message arguments may have to be stronger to produce the same degree of acceptance. Likewise, as a message becomes more proattitudinal, the message arguments may have to be weaker to produce the same degree of message rejection.

A third factor to consider about personal relevance is that in the "real world," there is likely to be a natural confounding between the personal relevance of an issue and the amount of prior thinking a person has done about the pool of issue-relevant arguments. There are at least two potentially important consequences of this prior thinking. First, because of the prior consideration, people may have a greater ability or may be more practiced in defending their beliefs. This would reduce susceptibility to influence. Second, if a person has considered an issue many times in the past, it may be more difficult to motivate the person to think about another message on the same topic because the person may feel that all arguments have been evaluated (and rejected) already. This would make it less likely that new compelling arguments would be accepted.

A final factor to consider is the empirically derived nature of the strong and weak arguments used in our research. This empirical derivation is an important methodological tool in that it allows us to test the extent of argument processing induced by different variables (see Chapter 2). However, in the "real world," where persuaders are often confined to posing arguments that are veridical (rather than plausible), it may generally be difficult to generate arguments on some issues that elicit primarily favorable thoughts when people scrutinize them. Importantly, even if all of these factors combine to make it generally more difficult to obtain increased persuasion on counterattitudinal issues with increased personal relevance in the real world, the ELM accounts for this resistance by tracking the extent to which enhancing relevance affects the elaboration of the issue-relevant arguments presented.

#### *Issue Involvement vs. Response Involvement*

The variable of "issue involvement" or personal relevance with which we have been concerned can be contrasted with another kind of self-relevance referred to as "response involvement" (Zimbardo, 1960) or "task involvement" (Sherif & Hovland, 1961). In this second kind of involvement, the attitudinal issue per se is not particularly important or relevant to the person, but adopting a position that will maximize the immediate situational rewards is. For example, the issue of raising taxes in the U.S. has

personal implications for most U.S. taxpayers (high issue-involvement), whereas the issue of raising taxes in England does not. However, one's expressed attitude on the latter topic may become important while entertaining one's British boss for dinner (high response involvement; cf., Schlenker, 1980). In some cases, response involvement should lead to increased influence (e.g., Zimbardo, 1960) and in other cases to decreased influence (e.g., Freedman, 1964), depending upon which position the rewards favor.

Chaiken (1980) has argued and provided some evidence for the view that just as issue involvement can lead to enhanced message processing, so too can response involvement, such as whether or not a person expects to be interviewed on an issue. We suspect that to the extent that this is true, it would occur mostly when issue involvement is moderate. If issue relevance was very low, but response involvement was very high, impression management concerns (rather than concerns about adopting a veridical position based on examination of issue-relevant arguments) may determine the attitude expressed (see Cialdini, Levy, Herman, Kozlowski, & Petty, 1976, for evidence). Although it is possible for impression management concerns to lead to extensive issue-relevant cognitive activity in some situations (e.g., a student assigned to argue in a public debate may carefully research the position in order to make a favorable impression), more typically impression management concerns will not produce a systematic evaluation of issue-relevant arguments (Cialdini & Petty, 1981).

If high response involvement was introduced into a situation in which issue involvement was also very high, it would be difficult to enhance information processing further. In fact, Leippe and Elkin (in press) have provided evidence for the view that response involvement may actually *reduce* argument processing in these situations. In their experiment, subjects heard a message containing strong or weak arguments advocating either the institution of senior comprehensive exams (messages adapted from Petty, Harkins, & Williams, 1980) or the imposition of a campus parking fee (messages designed specifically for this study). Since the same effects were observed for both topics, this variation will not be discussed further. Both issue and response involvement were manipulated in the study. In the high issue involvement conditions, subjects were told that the exams or parking fees were being considered for institution the following year, whereas in the low relevance conditions, subjects were told that the policy change was being considered for six years in the future (cf., Apsler & Sears, 1968; Petty & Cacioppo, 1979b). In the high response involvement conditions, subjects were told that following the message they would discuss their attitudes with another student and would be interviewed about the discussion by a psychology professor, whereas in the low response involvement conditions they were told nothing about the discussion or interview (cf., Chaiken, 1980, Experiment 1; Cialdini, Levy, Herman, Kozlowski, & Petty 1976).

The attitude results from the Leippe and Elkin (in press) study are

depicted in Figure 4-3. In addition to a main effect for argument quality, two two-way interactions and a three-way interaction were obtained. An Issue involvement  $\times$  Argument quality interaction replicated the results for issue relevance that we reported previously (i.e., Petty & Cacioppo, 1979b, see Figure 4-2). Increased issue relevance enhanced message processing such that high relevance subjects were more responsive to message quality than low relevance subjects. Interestingly, the Response involvement  $\times$  Argument quality interaction indicated that increased response involvement *reduced* message processing such that high response involvement subjects were less attentive to argument quality than low response involvement subjects. Finally a three-way interaction (see Figure 4-3) indicated that response involvement disrupted processing mostly when issue relevance was high.

In sum, it is now clear that increased issue relevance enhances processing of the arguments in a persuasive communication (see also Burnkrant & Sawyer, 1983). However, this effect appears most likely when people are free of impression management concerns. When response involvement is high, people may be distracted from processing the message arguments owing to anxiety about self-presentation. Alternatively, people may still engage in message processing under high response involvement conditions (if issue involvement is also high), but may simply *report* less polarized attitudes because these more cautious attitudes may be easier to defend in public (cf., Cialdini & Petty, 1981; Leippe & Elkin, in press). Future research that includes both public and private measures of opinion may untangle these possibilities.

#### Effects of Personal Responsibility on Message Elaboration and Persuasion

We have argued and provided evidence for the view that personal relevance (issue involvement) enhances motivation to process issue relevant arguments. There is also reason to believe that personal responsibility produces similar effects. Ever since Ringelmann found that group productivity on a rope-pulling task failed to reach the levels predicted based on individual performance (see Steiner, 1972), several contemporary social psychologists have replicated this effect and pursued its underlying cause. Recent research has documented that at least part of the reduced performance in groups (called "social loafing" by Latane, Williams, & Harkins, 1979) results from loss of motivation rather than ability (Ingham, Levinger, Graves, & Peckham, 1974; Latane et al., 1979). In addition to rope pulling, the reticence to exert physical effort in groups has been shown on a wide variety of tasks including helping another person in distress (Latané & Darley, 1970), picking up coins in an elevator (Latané & Dabbs, 1975), clapping one's hands (Harkins, Latane, & Williams, 1980), using a hand pump (Kerr &

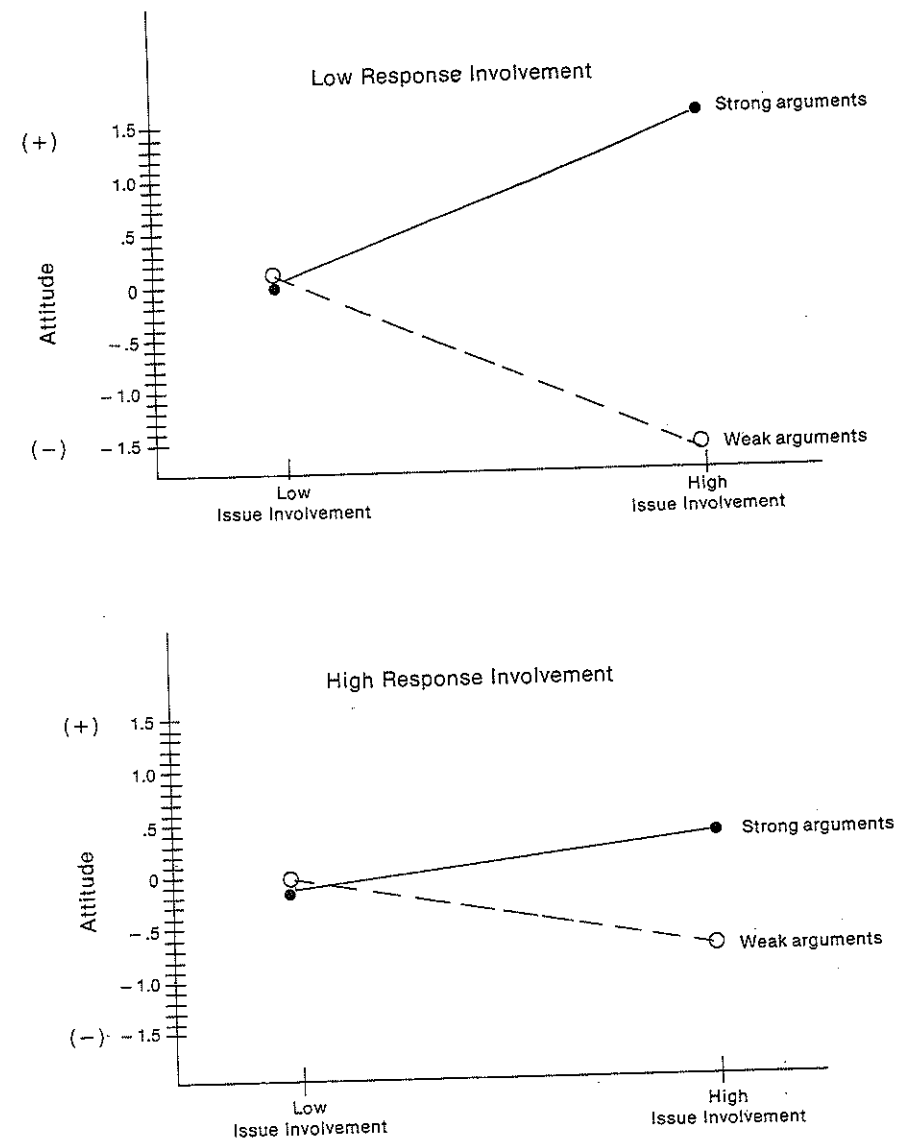


Figure 4-3. Postmessage attitudes as a function of the personal relevance of the issue, response involvement, and argument quality (data from Leippe & Elkin, in press).



Bruun, 1981), shouting out loud (Williams, Harkins, & Latane, 1981), fastening nuts and bolts (Petty, Cacioppo, & Kasmer, 1985), and others.

#### *Personal Responsibility Enhances Cognitive Work*

Although most of the research following Ringlemann has focused on tasks requiring physical exertion (e.g., Harkins, Latane, & Williams, 1980; Kerr & Bruun, 1981), in an exploratory study we examined the possibility that people who shared responsibility for a *cognitive* task would exert less *mental* effort than people who were individually responsible (Petty, Harkins, Williams, & Latane, 1977). Students were led to believe that they would either be the only person evaluating a communication, one of four, or one of 16 evaluators. We predicted that to the extent that cognitive effort is seen as costly and others are available to share the load, individual members of a group may be tempted to reduce their own efforts.

The college student subjects in this study were told that the journalism school at their university was evaluating its programs and needed feedback on the writing abilities of its students. The subjects were given booklets containing an editorial and a poem, rating scales, and written instructions informing them that, "Your task is to critically evaluate the poem (editorial). You are the only (one of four) (one of 16) person(s) who will be reading this particular poem (editorial). Thus, you alone (the four of you) (the 16 of you) have (share) the full responsibility for the critical evaluation." In addition, group evaluators were instructed that, "Your reactions will be combined with those of the other three (15) persons to form one overall rating for the poem (editorial) all of you read."

All subjects actually read and rated the same two communications; a short poem on life's unanswered questions and a two-page editorial containing a number of arguments favoring a reduction in tuition. After exposure to each stimulus, subjects were asked three questions designed to measure their perceived cognitive involvement in the task (e.g., to what extent were you trying hard to evaluate the communication?), and three evaluative questions about the material (e.g., to what extent do you feel the communication made its point effectively?). Consistent with the view that shared responsibility reduces cognitive work, subjects who were individually responsible for evaluating the communications reported putting more effort into their evaluations than those who worked in groups. In this initial investigation, the communications were not pilot tested; however, subjects were found to rate both the editorial and the poem favorably. Moreover, students who were solely responsible for the evaluation rated the communications more positively than those who shared responsibility.

#### *Additional Evidence for the Link between Responsibility and Cognitive Work*

Although only self-report measures of cognitive work were obtained in our initial study of group responsibility and cognitive effort, subsequent research has obtained convergent evidence. For example, Harkins and Petty (1982) employed a brainstorming task in which students were asked to

generate uses for objects. The students were either told that "you alone are responsible for listing uses" or that "you share the responsibility for listing uses for this object with nine other persons whose uses will be combined with yours." When confronted with objects for which it was relatively easy to generate uses (i.e., knife, box), solely responsible subjects generated significantly more uses than subjects who shared the responsibility.

In three studies, Brickner, Harkins, & Ostrom (in press) explored social loafing on an attitudinal task. Student subjects were asked to list their thoughts about the implementation of senior comprehensive exams (no messages were presented). Subjects were either told that they were the only person listing thoughts or that they shared the responsibility with a partner. In addition, the personal relevance of the exam proposal was varied by telling subjects either that the exam proposal was being considered for next year at their own university or that it was being considered either for a future date or for another university. When the issue was low in personal relevance, subjects who shared responsibility generated significantly fewer thoughts than those who were individually responsible. As might be expected if personal relevance motivates issue-relevant thinking (Petty & Cacioppo, 1979b), less loafing occurred in groups when the issue had high personal relevance.

Finally, Weldon and Gargano (1985) asked students to give an overall evaluation of various part-time jobs that were described on five different dimensions. The subjects were told either that they were the only person responsible for evaluating the jobs or one of 16 (whose ratings would be averaged). In their first experiment, subjects were told to evaluate as many jobs as possible in the time allotted. Subjects who were solely responsible evaluated significantly more jobs than those who shared responsibility. In their second experiment, all subjects were required to evaluate the same number of jobs. Interestingly, however, regression analyses indicated that subjects who were solely responsible used more of the available dimensions in forming their job ratings than subjects who shared responsibility. In sum, increasing personal responsibility for a task appears to affect both the quantity and quality of the cognitive effort expended.

#### *Personal Responsibility Enhances Message Elaboration*

The implications of the research on cognitive effort and responsibility for persuasion are straightforward: The greater the personal responsibility for evaluating an issue, the more people should be willing to exert the cognitive effort necessary to evaluate the issue-relevant arguments presented. Hence, individual evaluators are expected to regard a strong version of a message more favorably and a weak version of a message less favorably than group evaluators. To test this hypothesis, we asked undergraduates to provide peer feedback on strong or weak versions of editorial messages ostensibly written by journalism students (Petty, Harkins, & Williams, 1980, Experiment 2). It should be noted that at least three other social psychological models—

cognitive dissonance theory (Festinger, 1957), commodity theory (Brock, 1968), and deindividuation theory (Zimbardo, 1970), make a constrasting prediction in this study: individuals should provide more favorable evaluations than group evaluators, regardless of argument quality (see Petty, Cacioppo, & Harkins, 1983, for further discussion).

**Method.** A total of 180 students participated in a 2 (Group size: 1 or 10)  $\times$  3 (Argument quality: strong, mixed, or weak) between-subjects factorial design. Subjects were led to believe that they were either the only person responsible for evaluating an editorial or one of ten people who shared the responsibility. Subjects received one of three versions of a message arguing that seniors should be required to pass a comprehensive exam in their major as a requirement for graduation. Pilot testing indicated that one message contained strong arguments, another contained weak arguments, and a third contained a mixture of arguments (and elicited a mixture of favorable and unfavorable thoughts). After reading the appropriate message, subjects provided an evaluation of the message, and listed their thoughts.

**Results.** Conceptually replicating the results of the Petty, Harkins, Williams, and Latane (1977) study, subjects who believed they were the only evaluator of the editorial reported expending more cognitive effort than did subjects who believed they were one of ten evaluators. The attitude results, graphed in Figure 4-4, revealed an Argument quality  $\times$  Responsibility interaction. As personal responsibility for evaluation decreased, the quality of the arguments in the message became a less important determinant of the evaluations. More specifically, group evaluators were significantly more favorable toward the weak message, but were significantly less favorable toward the strong message than individual evaluators. As expected, evaluations of the mixed message were unaffected by the extent of responsibility. Analyses of the listed thoughts provided further support for the hypothesis. Increasing the number of evaluators reduced the number of unfavorable thoughts elicited by the weak message and reduced the number of favorable thoughts elicited by the strong message. The group size manipulation had no effect on the profile of thoughts elicited by the mixed message.

In sum, subjects who were part of a group that was responsible for an evaluation generated fewer favorable thoughts about the strong version of the editorial than did subjects who were solely responsible for the evaluation and fewer unfavorable thoughts about the weak version. Thus, in addition to replicating the results of our initial study on the effects of personal responsibility on perceived cognitive effort (Petty, Harkins, Williams, & Latane, 1977), this experiment demonstrated that group evaluators actually generated fewer thoughts consistent with the quality of

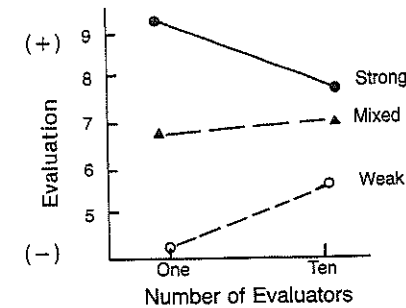


Figure 4-4. Postmessage evaluations as a function of the perceived number of message evaluators and argument quality (data from Petty, Harkins, & Williams, 1980, Experiment 2).

the message arguments than did individual evaluators. In addition, group evaluators rated the weak version more favorably, and the strong version less favorably, than did individual evaluators. When the stimulus to be evaluated was neither very strong nor very weak, group size did not produce significant effects on thoughts or evaluations. Finally, within-cell correlations revealed that the more perceived effort expended, the more favorable the evaluation of the strong version of the editorial, but the more negative the evaluation of the weak version of the editorial. The extra message elaboration characterizing individual evaluators apparently renders them better able to discover and appreciate the virtues of a strong appeal and the flaws of a weak one.

One important area of potential applicability of the research on group cognitive responsibility is to jury decision making. The work on group responsibility and effort suggests that as jury size increases people may be less motivated to participate in the group. Kessler (1973) examined the participation rates of mock jurors in 6 and 12 member groups and found evidence consistent with this speculation. Her results revealed that although only 4% of jurors in 6-member juries were completely silent, 25% of jurors in 12-member juries said nothing. One direct implication of the reduced motivation to process issue-relevant arguments in groups is that small juries should be more sympathetic to defendants than large juries when the evidence against the defendant is weak, because small (more responsible) juries will better realize how flawed the testimony is. On the other hand, when the evidence against the defendant is strong, a large (less individually responsible) jury should be preferred by the defendant (though not society) since it will be less likely to realize the cogency of the damaging testimony. Results consistent with this hypothesis were obtained in a study of mock juries by Valenti and Downing (1975). When the evidence against a defendant was weak, 80% of 6-member juries and 60% of 12-member juries



voted for acquittal; when the evidence against the defendant was strong, however, 90% of 6-member juries voted for conviction, whereas only 20% of 12-member juries did so.

#### Effects of Number of Message Sources on Message Elaboration and Persuasion

In the previous section we provided evidence for the view that increasing the number of message recipients responsible for message evaluation could reduce motivation to process an advocacy. In this section we provide evidence for the view that increasing the number of message sources presenting an advocacy can enhance motivation to process a message. There are number of everyday situations in which persuasive information is provided by several sources rather than just one. For example, at political rallies, a number of speakers argue in favor of or against a candidate or some issue position. In advertising, multiple spokespersons provided testimonials for a product. In trials, several witnesses provide character evidence about a defendant. Despite the frequency of occurrence of persuasion by multiple sources, relatively little is known about the underlying processes mediating the effectiveness of this strategy (Harkins & Petty, 1983).

Traditional analyses of the number of message sources have assumed that the more people who are perceived to advocate a position (up to some limit), the more conformity pressure that is induced, and the more agreement that results (e.g., Asch, 1951; Latane & Wolf, 1981; White, 1975). One popular explanation for this conformity effect is that people shift toward the majority view out of a desire to hold a correct opinion (Festinger, 1954). The more people who endorse a position, the more correct it presumably is. However, Burnstein and Vinokur and their colleagues (e.g., Burnstein & Sentsis, 1981; Burnstein & Vinokur, 1977) have suggested that an alternative process underlies the effects of multiple sources. Specifically, they have argued that when people are confronted by the opinions of others, they are motivated to figure out *why* the others have advocated these views. According to this analysis, the more other people who advocate a position, the greater is the motivation to consider possible reasons behind the proposal. This curiosity can result in people self-generating arguments in favor of the position advocated. Both the "conformity" and the "persuasive arguments" explanations are relevant to situations in which no arguments are presented by the sources, and both theories predict that multiple sources should be associated with enhanced agreement. The conformity explanation postulates attitude change via the peripheral route (a simple inference about correctness is sufficient to induce change), whereas the persuasive arguments explanation postulates attitude change via the central route (change is a result of effortful issue-relevant argument generation). Harkins and Petty (1981a) proposed a variation of the "persuasive arguments"

hypothesis to account for situations in which a varying number of other people actually presented arguments to message recipients. Specifically, it was argued that the more people who presented the arguments, the more scrutiny each argument would receive.

#### *Initial Evidence for the Multiple Source Effect*

In an initial study designed to explore the information processing effects of multiple sources, Harkins and Petty (1981a, Experiment 1) exposed students to one or three arguments presented by one or three sources. Subjects were told that a faculty committee had videotaped three fellow students giving their views on a recent proposal to implement senior comprehensive exams. The subjects were further told that each of the students favored the comprehensive exam idea and that all were instructed to provide three arguments in favor of their positions. Subjects were led to believe that they would be exposed to a randomly selected segment of the tapes made so far. Subjects in fact saw either: (a) one source giving one argument in favor of the exams, (b) one source giving three arguments, (c) three sources giving the same basic argument, or (d) three sources each giving three different arguments. Two control groups were also incorporated into the design. A "background-only" control group was exposed only to the background information that three sources had each given three arguments in favor of the exams, but they were not actually exposed to any sources or arguments. An "attitude-only" control group completed the attitude measures without exposure to the background information, sources, or arguments. These control groups allowed a determination of whether actual exposure to multiple sources and arguments had persuasive impact on message recipients over and above that achieved by the mere knowledge of the number of sources and arguments to which they might be exposed.

Analysis of attitudes toward senior comprehensive exams revealed that "background-only" control subjects expressed more favorable attitudes than those held by the "attitude-only" subjects. Thus, simple knowledge of the number of sources and arguments in favor of a position was sufficient to produce attitude change. It was not clear, of course, whether this change was due to the simple "conformity" (peripheral route) or "persuasive arguments" (central route) explanation because subjects may have changed as a result of simple inferences based on the background information, or may have actively attempted to generate arguments in favor of senior comprehensive exams as a result of exposure to the background information (Burnstein & Sentsis, 1981). Importantly, however, subjects exposed to multiple sources presenting multiple arguments showed significantly more agreement with the exam proposal than subjects in both control groups.

The fact that subjects exposed to multiple sources presenting multiple arguments showed more agreement than subjects in any other condition is consistent with two plausible explanations. The first is an *attributional* hypothesis. Participants seeing three different sources each generate a different argument might reasonably conclude that a large pool of good

arguments in favor of the advocated position must exist, and change as a result of this simple inference (peripheral route). Subjects in the other cells would be less likely to make this inference. A second explanation is based on *message elaboration*. This account suggests that participants seeing multiple sources present multiple arguments engage in more argument-relevant processing than participants in the other conditions. That is, each time a novel person appears, the participant is motivated to hear what this new source has to say. If the arguments are new, this enhanced processing may result in the generation of additional thoughts favorable to the advocacy. If the arguments are old, however, the enhanced processing might result in no *new* favorable thoughts, especially if the repeated argument was sufficiently simple as to have been processed completely during the initial exposure (see Chapter 3 discussion of message repetition; Cacioppo & Petty, 1985). If the same source appears again, even though with new arguments, less effort may be put into processing because the recipient feels that this source has been heard from already. Consistent with this reasoning, we found that subjects receiving multiple arguments from multiple sources generated more favorable thoughts about senior comprehensive exams than subjects in any other cell.

The attributional and elaboration interpretations were compared in a second study (Harkins & Petty, 1981a, Experiment 2). In a key condition of this study, subjects receiving multiple arguments from multiple sources were told that the arguments that were presented exhausted the pool of good arguments favoring the position. In this condition, the attributional interpretation would predict no persuasive advantage resulting from multiple sources presenting multiple arguments, since the argument pool was limited to the number of arguments presented. Although manipulation checks revealed that the argument pool induction was successful, limiting the pool of arguments did not reduce persuasion. Multiple sources presenting multiple strong arguments were still more persuasive than when the same strong arguments were presented by one source.

#### *Testing the Elaboration-Enhancement Hypothesis*

In order to provide more definitive evidence for the view that multiple sources enhance message processing, and following the logic presented in Chapter 2, a third study was conducted in which the number of message sources was manipulated along with the quality of the issue-relevant arguments supporting the advocacy (Harkins & Petty, 1981a, Experiment 3). If multiple sources enhance argument processing, then strong arguments should become more persuasive when presented by multiple sources, but weak arguments should become less persuasive when presented by multiple sources.

*Method.* A total of 100 students was randomly assigned to the cells of a 2 (Number of sources: one or three)  $\times$  2 (Argument quality: strong or weak)

between-subjects factorial design. Subjects were told that the Faculty Committee on Academic Affairs had solicited students' written opinions on the issue of instituting senior comprehensive exams. All subjects were told that the opinions of three students had been obtained, and so far all the students favored the exam proposal. Subjects were told that each student had written three arguments in favor of the proposal, and that they would examine a random selection from the materials collected. The subjects were given an envelope containing the written materials. Each argument appeared on a separate page and before each argument a statement attributed the message to one of three male students: "Here is an argument generated by (one of three male names), who is a Junior." If the subject was in the one-source condition, the statement preceding the remaining arguments said: "Here is another argument generated by (same name as before), who is a Junior." In the multiple source conditions, the statement preceding the remaining arguments said: "Here is an argument generated by another student, (different name), who is a Junior." Which of the three names was used, with what arguments the names were paired, and in what order they were presented was randomly determined for each subject. In sum, the subjects were exposed to either three strong or three weak arguments presented by one or three sources. Following examination of the written materials, subjects responded to attitude, thought-listing, and ancillary measures.

*Results.* Subjects exposed to three writers who generated weak arguments felt that fewer of their classmates would support the proposal than one-source subjects, but there were no differences for the subjects exposed to high quality arguments. Also, there were no differences among conditions in the number of good arguments that subjects believed existed in support of the proposal.

Importantly, an analysis of the measure of attitudes toward senior comprehensive exams revealed a Number of sources  $\times$  Argument quality interaction (see Figure 4-5): subjects exposed to three persons presenting strong arguments were more persuaded and subjects exposed to three persons presenting weak arguments were less persuaded than subjects exposed to exactly the same information presented by just one person. The results for the thought generation measure followed the same pattern. Three-source subjects exposed to the strong arguments generated more favorable thoughts and tended to generate fewer unfavorable thoughts than one-source subjects exposed to the same arguments. On the other hand, three-source subjects exposed to the weak arguments generated more unfavorable thoughts and fewer favorable thoughts than one-source subjects exposed to the same information.

Consistent with the elaboration-enhancement hypothesis, then, increasing the number of message sources intensified information-processing activity. Further support for the view that multiple sources affect persuasion

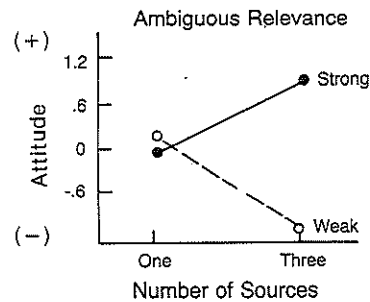


Figure 4-5. Postmessage attitudes as a function of the perceived number of message sources and argument quality (data from Harkins, & Petty, 1981a, Experiment 3).

by increasing issue-relevant elaboration comes from an additional study (Harkins & Petty, 1981b) in which it was found that having subjects perform a secondary distraction task eliminated the multiple source effect without affecting recall, the number of good arguments thought to exist in favor of the advocacy, or the percentage of classmates thought to support the proposal. When no distraction was present, the multiple source effect was replicated. In short, this study suggests that distraction appears to disrupt the additional processing elicited by multiple sources (see discussion of distraction in Chapter 3).

#### *Why Do Multiple Sources Enhance Elaboration?*

The evidence we have just reviewed is very consistent with the view that people think more about information that comes from multiple sources than the same information presented by a single individual. However, it is not clear what it is about multiple sources that leads to the additional processing. One possibility is that subjects may reason that it is more worthwhile to think about information from multiple sources because this information, compared with that generated by one source, is more likely to be based on different perspectives and independent pools of knowledge. If arguments from multiple sources are perceived as representing independent perspectives on an issue, then this information may be more worthy of diligent consideration than information from only one perspective.

Harkins and Petty (in press) report three studies that are consistent with this analysis. In one experiment, students were exposed to three strong arguments in favor of senior comprehensive exams that were presented by either one source, three independent sources, or three sources that were made nonindependent by telling subjects that the sources belonged to a committee that had worked together in generating their arguments (cf., Wilder, 1977). Multiple independent sources produced more agreement than when the same information was presented by a single source (replicating the research described above), but multiple sources described as belonging to a committee did not.

In a second study, subjects were informed either before or after argument exposure that the multiple sources had formed a committee. When the committee information preceded argument exposure, the persuasive advantage of multiple sources was lost. However, when this information came after message exposure, the multiple source effect remained. When subjects learn of the nonindependence of the sources after message exposure, it is too late to affect argument processing.

Finally, in a third study, subjects were led to believe that the multiple sources who had formed a four-member committee were either very similar or very different from each other. In the similar perspectives condition, subjects were told that the committee members had been selected to be as similar as possible. Also, in introducing the committee members, each was described as having the same major (either English, Education, Marketing, or Physics). In the divergent perspectives condition, subjects were told that the committee members had been selected to be as different as possible. In addition, all of the committee members were introduced as having different majors. When the different perspectives of the committee members were made salient, multiple sources presenting strong arguments were more persuasive than a single source presenting the same information. However, multiple similar sources who had formed a committee were no more persuasive than a single source. These studies are consistent with the view that the power of multiple sources to enhance elaboration lies in their perceived informational independence and in the divergent perspectives they are presumed to represent.<sup>2</sup>

#### *Effects of Need for Cognition on Message Elaboration and Persuasion*

##### *Need for Cognition Enhances Cognitive Work and Message Evaluation*

In Chapter 3 and in the preceding sections of this chapter, we have suggested that situational factors can play an important role in enabling and motivating individuals to think extensively versus superficially about the merits of an appeal. We reasoned that just as there are situational factors such as distraction, issue relevance, personal responsibility, and number of message sources that influence the likelihood that individuals will think about and elaborate upon the externally provided message arguments, so too must there be dispositional factors governing message processing and, indirectly, persuasion. As noted in Chapter 2, we proposed a functional relationship between an individual's level of need for cognition and his or her tendency to extract and cognitively elaborate on information from a persuasive communication. Our first experimental test of this hypothesis was designed to be a conceptual replication of the Petty, Harkins, and Williams (1980) study on group responsibility, with the major difference

<sup>2</sup>In Chapter 8 we address the issue of when multiple sources could serve as peripheral cues according to the ELM.

being that we examined the role of a dispositional (need for cognition) rather than a situational (group size) factor on cognitive effort and message evaluation (Cacioppo, Petty, & Morris, 1983, Experiment 1).

**Method.** A group of 527 undergraduate students in introductory psychology classes participated in a preliminary testing session during which they completed a battery of tests. Included in the battery was the need for cognition scale (NCS; Cacioppo & Petty, 1982) and an attitude survey regarding campus issues. Students were classified as being low, moderate, or high in need for cognition based on their responses to the NCS. Pairs of students who fell in the bottom or the top tripartite of scores and who reported approximately the same attitude toward instituting senior comprehensive exams were recruited for participation in the experiment. Students did not know what facet of the preliminary testing was responsible for their eligibility, and the experimenter did not know the subjects' levels of need for cognition when they participated in the experiment. Ultimately, 114 undergraduates participated in a 2 (Need for cognition: high, low)  $\times$  2 (Argument quality: strong, weak) between-subjects factorial design.<sup>3</sup>

At the beginning of the session, subjects were instructed that their university branch of the (fictitious) National Society of Journalism Teachers was undergoing an evaluation of its program and that the psychology department was assisting by studying the subjective impact of various journalistic styles. The subjects were further told that that would be reading an editorial prepared by a journalism student for possible broadcast and publication. Subjects then read either a strong or weak set of arguments favoring the recommendation that senior comprehensive exams be instituted as a prerequisite for graduation at their university. Afterward, subjects indicated their evaluation of the communication, rated how much cognitive effort they exerted in evaluating the message, tried to recall the message

<sup>3</sup>As noted in Chapter 2, argument quality has been defined operationally such that "strong" arguments elicit more favorable than unfavorable thoughts about the recommendation, whereas "weak" arguments elicit more unfavorable than favorable thoughts about the recommendation. In pretests for this study, we observed that a "strong" message that we had used previously again tended to elicit more favorable than unfavorable thoughts when need for cognition was ignored. However, when the profile of thought-listings was examined within each NC group, we found that people high in NC were actually generating slightly more unfavorable thoughts (e.g., counterarguments) than favorable thoughts to the "strong" message. Perhaps because of their greater tendency to think about incoming information, individuals high in NC access more prior knowledge and/or a more developed attitude schema. If so, it should be more difficult to generate what individuals high, in contrast to low, in NC respond to as strong arguments on a counterattitudinal issue. In any case, modifying the strong arguments proved sufficient to elicit primarily favorable thoughts from both individuals high and low in NC, but these observations underscore the importance of equating conditions for such factors when using the individual differences approach for theory testing.

arguments, expressed their impressions of the communicator, and completed several ancillary measures.

**Results.** Cell means for message evaluation are depicted in Figure 4-6. As expected, the crucial Need for cognition  $\times$  Argument quality interaction appeared. Pairwise comparisons revealed that individuals high in need for cognition provided more discriminating judgments of the externally provided message arguments than did individuals low in need for cognition. The analysis of the recall data revealed that individuals high in need for cognition recalled substantially more message arguments ( $M = 3.98$ ) than their counterparts ( $M = 2.91$ ). Moreover, analysis of the measure of cognitive effort revealed that subjects high in need for cognition reported expending more effort thinking about the editorial ( $M = 5.93$ ) than did subjects low in need for cognition ( $M = 4.98$ ). A main effect for argument quality signified that subjects reported thinking more about the weak than the strong message arguments, but a Need for cognition  $\times$  Argument quality interaction indicated that this effect primarily characterized subjects high in need for cognition.

#### *Additional Evidence for the Link between Need for Cognition, Message Elaboration, and Persuasion*

The data from the first experiment support the notion that individuals high in need for cognition are inclined to extract more from and think more about the message arguments than individuals low in need for cognition. A second experiment was conducted, however, to replicate and extend these findings (Cacioppo, Petty & Morris, 1983, Experiment 2). Pretesting replicated previous surveys showing that students high and those low in NC did not differ in their attitudes toward raising student tuition at their university, so this topic was selected to avoid concerns about using

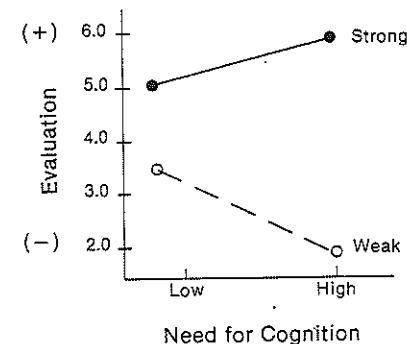


Figure 4-6. Postmessage evaluations as a function of need for cognition and argument quality (data from Cacioppo, Petty, & Morris, 1983 Experiment 1).

unrepresentative samples from the population of students high versus low in NC. Furthermore, postcommunication measures of attitudes were collected in addition to measures of message evaluation to determine whether the attitudes of individuals high, in contrast to low, in NC were more likely to be formed through effortful deliberations regarding the merits of the recommendation.

**Method.** A group of 110 female undergraduates participated in a 2 (Need for cognition: high or low)  $\times$  2 (Argument quality: strong or weak) between-subjects factorial. Subjects were informed that the Office of Student Academic Affairs at their university was reevaluating its policies and was seeking recommendations about possible changes. Subjects were told that policy statements had been prepared for possible broadcast and publication in the local media to inform and to obtain the reactions of people in the university community. Subjects were asked to read one of the policy statements, rate it for readability, and respond to the questions about the policy statement that followed it in their booklet.

Following this background material, subjects read an approximately 300-word message justifying the proposal that student tuition be increased at their university; half of the subjects read a set of strong message arguments, and half read a set of weak message arguments. Next, subjects completed scales to assess their attitudes, message evaluation, level of need for cognition, and other ancillary measures.

**Results.** Analysis of the measure of message evaluation yielded main effects for Argument quality and for Need for cognition that were qualified by the expected Need for cognition  $\times$  Argument quality interaction. Pairwise comparisons revealed that, as in the previous experiment, argument quality had a greater impact on the evaluations of subjects high than of those low in NC. This result again suggests that the former individuals scrutinized more the message arguments that were provided.

The analysis of the attitude index yielded a main effect for Argument quality and the expected Need for cognition  $\times$  Argument quality interaction (see Figure 4-7). Importantly, the pattern of means on the attitude measure was similar, but not identical, to that found for message evaluation ( $r = 0.52$ ). We reasoned that if subjects high in need for cognition were more likely to derive their attitude through a considered evaluation of the arguments central to the recommendation, then there should be a stronger association between message evaluations and attitudes in subjects high than in those low in NC. To test this reasoning, a separate correlation was calculated between these measures in the high and the low NC groups. As expected, this correlation was significantly larger in the HNC group ( $r = 0.70$ ) than in the LNC group ( $r = 0.22$ ).

To further test the notion that it was the deliberation about the merits of the message arguments that led to the observed attitudes rather than vice

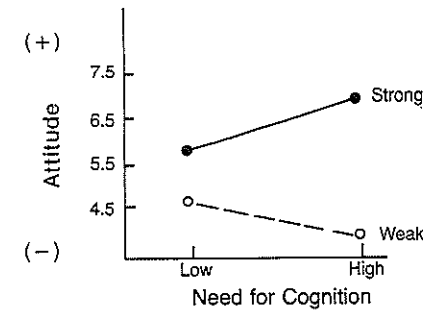


Figure 4-7. Postmessage attitudes as a function of need for cognition and argument quality (data from Cacioppo, Petty, & Morris, 1983, Experiment 2).

versa, we conducted two analyses of covariance (ANCOVAs). In the first, postcommunication attitude served as the criterion measure, and message evaluation served as the covariate; in the second, message evaluation served as the criterion, and postcommunication attitude served as the covariate. The results revealed that the Need for cognition  $\times$  Argument quality interaction for message evaluation remained significant when attitudes served as the covariate, whereas any hint of a significant Need for cognition  $\times$  Argument quality interaction on the attitude measure was eliminated when the measure of message evaluation served as the covariate. These results, too, are consistent with the view that need for cognition affects message processing and thereby affects people's susceptibility or resistance to persuasion.

#### *Individuals Low in Need for Cognition: Verbal Dolts or Cognitive Misers?*

Throughout this research, we have conceived need for cognition as acting primarily as a motivational factor—or more specifically, that need for cognition represents a person's level of intrinsic motivation to engage in and enjoy effortful cognitive endeavors. Although it is certainly reasonable that intelligent individuals are more likely to experience success and attain aspired goals in difficult cognitive tasks, positive reinforcement per se is not sufficient for the development of intrinsic motivation (e.g., Deci, 1975; Deci & Ryan, 1980). Hence, a strong correlation between need for cognition and intelligence would be expected only when dealing with very heterogeneous populations (e.g., assembly line workers vs. university faculty); and the tendency for individuals high in need for cognition to deliberate more about the merits of the arguments in a persuasive appeal should be obtained even when controlling for individual differences in intelligence.

Previous research is consistent with this reasoning. Cacioppo, Petty, and Morris (1983) administered the NCS and the Shipley-Hartford (1940) scale to 104 undergraduates drawn from introductory psychology courses and found that the need for cognition was unrelated to abstract reasoning ( $r =$

-0.03) and weakly and nonsignificantly related to verbal reasoning ( $r = 0.15$ ). In a subsequent replication, Morris, Bachman, Bromwell, and Sterling (1982) found need for cognition to be unrelated to abstract reasoning ( $r = 0.12$ ) and weakly related to verbal reasoning ( $r = 0.21$ ). Yet these results are not definitive. Hence, an experiment was conducted in which both need for cognition and intelligence were gauged to examine more rigorously whether the effect of need for cognition on message processing is best explained by the view that individuals low in need for cognition act as cognitive misers (motivational difference) or that they are verbal dolts (ability) (Cacioppo, Petty, Kao, & Rodriguez, in press, Experiment 1).

**Method.** A group of 185 undergraduates completed participation in a 2 (Need for cognition: high or low)  $\times$  2 (Argument quality: strong or weak) between-subjects factorial. When subjects arrived at the laboratory, they were told that the study bore on extrasensory communication. They were informed that their role in this study was to listen to an audiotaped message and to try to transmit their thoughts during the message to a "receiver" located in another room. To lend credence to the cover story, the experimenter recruited a volunteer to act as the receiver. After placing the receiver in a separate room, subjects read background information stating that the recommendation about which they would be hearing had been advanced recently by a University Committee on Academic Affairs and Policy Formation. They read that the committee was considering policies for the university in the 1990s. Subjects then listened to an audiotaped message in which the speaker was introduced and which consisted of either eight strong or eight weak arguments favoring the proposal that tuition be substantially increased in 1990—a proposal that presumably would not affect any of the subjects. Afterward, subjects completed scales designed to assess their attitudes, evaluation of the message, and cognitive effort expended. In addition, subjects completed the NCS and the Shipley-Hartford Vocabulary Test (Shipley, 1940), and they were given 5 min to recall message arguments. It might be noted that a number of studies have found that correlations between the Shipley-Hartford scale and other assessments of intellectual functioning are quite high (e.g., Watson & Klett, 1968; Wiens & Banaka, 1960). Watson and Klett (1968), for example, reported a correlation of +0.78 between this scale and the Wechsler Adult Intelligence Scale (WAIS).

**Results.** Analyses of the indices of attitude and message evaluation replicated the Need for cognition  $\times$  Argument quality interactions observed in previous studies (see Table 4-1), and individuals high in need for cognition were again found to recall significantly more message arguments ( $M = 4.44$ ) than those low in need for cognition ( $M = 3.72$ ). The correlation between the measures of need for cognition and verbal intelligence, on the

Table 4-1. Message Evaluation and Postcommunication Attitude as a Function of Need for Cognition and Argument Quality

| Measure            | Low Need for Cognition |                   | High Need for Cognition |                   |
|--------------------|------------------------|-------------------|-------------------------|-------------------|
|                    | Weak Arguments         | Strong Arguments  | Weak Arguments          | Strong Arguments  |
| Message Evaluation | 5.37 <sup>b</sup>      | 6.29 <sup>c</sup> | 4.38 <sup>a</sup>       | 6.20 <sup>c</sup> |
| Attitude           | 5.01 <sup>b</sup>      | 6.30 <sup>c</sup> | 4.04 <sup>a</sup>       | 6.62 <sup>c</sup> |

Note: Means in a given row with the same superscript are not significantly different at  $p < 0.05$  by the Newman-Keuls test. Data from Cacioppo, Petty, Kao, & Rodriguez, in press).

other hand, was stronger than observed previously ( $r = 0.32$ ), providing a rich context in which to examine our conceptualization.

To reiterate, the notion underlying this research is that individuals low in need for cognition are avoiding effortful analyses of the persuasive communication (cognitive misers) rather than scrutinizing but failing to differentiate cogent from specious message arguments (verbal dolts). Our examination proceeded along three fronts. First, the effects of need for cognition and argument quality on reported cognitive effort were determined. If individuals low, in contrast to high, in NC are trying unsuccessfully to evaluate the strong and weak message arguments, then the cognitive effort they report having expended in evaluating the message should be comparable. Results revealed instead that individuals high in need for cognition reported expending more cognitive effort than individuals low in need for cognition—replicating our initial study in this area.

Second, subjects were divided into high and low intelligence groups on the basis of a median split on their verbal intelligence distribution. Separate 2 (Verbal intelligence)  $\times$  2 (Argument quality) ANOVAs were performed to determine whether the effects of intelligence mirrored those of need for cognition. Results revealed that individuals high in verbal intelligence recalled more message arguments ( $M = 4.37$ ) than did individuals low in verbal intelligence ( $M = 3.75$ ), apparently mirroring the need for cognition analysis. However, we have not argued that intelligence and other ability factors have *no* effects on message processing, but rather that the effects of need for cognition are separable. To test this reasoning, a stepwise regression analysis was performed using the number of arguments recalled as the criterion, and the verbal intelligence and need for cognition scores as the predictors. Consistent with the conceptualization of need for cognition as reflecting intrinsic motivation to engage generally in effortful cognitive endeavors, both verbal intelligence *and* need for cognition were found to account for significant and distinct sources of variance in the number of arguments recalled by subjects.

Further evidence for the conceptualization of need for cognition and



intelligence as separable factors was provided by the remaining analyses. Recall that subjects high in NC reported expending more cognitive effort than subjects low in NC, yet there was no significant difference between people high versus low in verbal intelligence on this measure. Moreover, the Dispositional factor  $\times$  Argument quality interactions on the indices of attitude and message evaluation, which were significant when subjects were blocked in terms of their need for cognition scores, were *not* significant when subjects were blocked in terms of their verbal intelligence scores. Instead, a significant main effect for verbal intelligence on the measure of message evaluation indicated that intelligent subjects rated the message more negatively overall than did their less intelligent counterparts.

Third, to yet further explore the effects on message processing of the subjects' need for cognition without variations in verbal intelligence, ANCOVAs were conducted with need for cognition serving as the blocking variable and verbal intelligence serving as the covariate (cf., Insko, Turnbull, & Yandell, 1974, for a discussion of this procedure). Results revealed that the interaction between need for cognition and argument quality on the indices of attitude and message evaluation remained significant even when variations in subjects' verbal intelligence were statistically controlled. The main effect for need for cognition on recall and on reported cognitive effort also remained significant.

Together, these data indicate that need for cognition and verbal intelligence are accounting for at least partially independent components of message processing. Verbal intelligence was found to be superior to need for cognition in predicting message recall, suggesting both that people's comprehension of externally provided message arguments is strongly influenced by their level of verbal intelligence (Eagly & Warren, 1976) and that verbal intelligence was assessed validly in the present research. But consistent with previous research, the retention of the externally provided message arguments neither predicted overall postcommunication attitude scores ( $r = -0.09$ , n.s.) nor the postcommunication attitudes observed within the strong arguments condition ( $r = 0.01$ , n.s.) or the weak arguments condition ( $r = -0.18$ , n.s.) (cf., Cacioppo & Petty, 1979a; Greenwald, 1968; Insko, Lind, & LaTour, 1976; Petty, Cacioppo, & Goldman, 1981).

Moreover, none of the significant effects of need for cognition on message processing and attitudes were obtained when intelligence served as the blocking variable, and all of the former effects remained significant when variations in people's verbal intelligence were controlled statistically. Thus, the present research provides evidence for the view that individuals low in need for cognition think less about persuasive communications than individuals high in need for cognition, and that this difference is due to the relative likelihood that individuals low in need for cognition will try to avoid effortful cognitive analyses of the incoming information.

## Retrospective

In this chapter, we have reviewed evidence consistent with the proposition that some variables affect persuasion by modifying a person's motivation to elaborate issue-relevant arguments in a relatively objective manner. Thus, when the personal relevance of a message is increased, or people have more personal responsibility for evaluating the message, or the information is provided by multiple independent sources, people become more motivated to process the issue-relevant arguments presented. The net result is that strong arguments are accepted more when motivation to process is high but weak arguments are rejected more. In addition, the individual difference variable, need for cognition, was shown to be an important determinant of elaboration likelihood. People who dispositionally tend to enjoy thinking show greater differentiation of strong and weak arguments than people who do not characteristically enjoy thinking. These results held even when differences in intelligence were controlled. Now that we have addressed both ability (Chapter 3) and motivational (Chapter 4) variables that affect information processing in a relatively objective manner, we turn in the next chapter to ability and motivational variables that tend to bias issue-relevant elaboration.